

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****Product Identifier**

Perihalan Produk: **Bismuth Sulfite Agar Base**  
Product Description: **Bismuth Sulfite Agar Base**  
Cat No. : R452402

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

**Details of the supplier of the safety data sheet**

Company Thermo Scientific Microbiology Sdn Bhd  
No.6, Jalan TTC 6, Taman Teknologi Cheng,  
Cheng, 75250 Melaka, Malaysia  
+606 334 0975 .

Supplier Remel  
12076 Santa Fe Drive Lenexa,  
KS 66215 United States  
Telephone: 1-800-255-6730  
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E-mail address mbd-sds@thermofisher.com

**Emergency Telephone Number**

(603) 5122 8888  
CHEMTREC Malaysia **1-800-815-308** (Malay)  
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION****Classification of the substance or mixture**

|                                   |                   |
|-----------------------------------|-------------------|
| Skin Corrosion/Irritation         | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation | Category 2 (H319) |

**Label Elements****Signal Word****Warning****Hazard Statements**

H315 - Causes skin irritation

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H319 - Causes serious eye irritation

## Precautionary Statements

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                   | CAS No      | Weight % |
|-----------------------------|-------------|----------|
| Ferrous sulfate             | 7720-78-7   | 0.58     |
| Ammonium hydroxide          | 1336-21-6   | 1.48     |
| C.I. Basic green 1          | 633-03-4    | 0.03     |
| Peptones, connective tissue | 102506-13-8 | 17.37    |
| Glucose                     | 50-99-7     | 13.51    |
| Agar                        | 9002-18-0   | 28.95    |
| Sodium phosphate dibasic    | 7558-79-4   | 7.72     |
| Sodium sulfite              | 7757-83-7   | 10.61    |
| Sodium carbonate            | 497-19-8    | 0.1      |
| Ferric ammonium citrate     | 1185-57-5   | 0.48     |
| Bismuth citrate             | 813-93-4    | 0.06     |
| Beef Extract                | 00-0-000    | 17.37    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

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## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

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## Control Parameters

| Component               | Malaysia | ACGIH TLV                | OSHA PEL                           |
|-------------------------|----------|--------------------------|------------------------------------|
| Ferrous sulfate         |          | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup> |
| Ferric ammonium citrate |          | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup> |

| Component               | European Union | The United Kingdom                                                | Germany |
|-------------------------|----------------|-------------------------------------------------------------------|---------|
| Ferrous sulfate         |                | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr |         |
| Ferric ammonium citrate |                | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr |         |

## Exposure Controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

### Eye Protection

Goggles

### Hand Protection

Protective gloves

### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

#### Physical State

Powder Solid

#### Odor

No information available

#### Odor Threshold

No data available

#### pH

No information available

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

No information available

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Flash Point                             | No information available | Method - No information available |
| Evaporation Rate                        | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Specific Gravity / Density              | No data available        |                                   |
| Bulk Density                            | No data available        |                                   |
| Water Solubility                        | No information available |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| C.I. Basic green 1                      | 0.765                    |                                   |
| Sodium sulfite                          | -4                       |                                   |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | Not applicable           | Solid                             |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

#### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat. Avoid dust formation.

### Incompatible Materials

None known.

### Hazardous Decomposition Products

None under normal use conditions.

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## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Acute Toxicity

#### Toxicology data for the components

| Component                | LD50 Oral                | LD50 Dermal           | LC50 Inhalation                               |
|--------------------------|--------------------------|-----------------------|-----------------------------------------------|
| Ferrous sulfate          | LD50 = 319 mg/kg ( Rat ) |                       |                                               |
| Ammonium hydroxide       | LD50 > 350 mg/kg (Rat)   |                       |                                               |
| Glucose                  | 25.8 g/kg ( Rat )        |                       |                                               |
| Agar                     | LD50 = 11 g/kg ( Rat )   |                       |                                               |
| Sodium phosphate dibasic | LD50 = 17 g/kg ( Rat )   |                       |                                               |
| Sodium sulfite           | 2610 mg/kg ( Rat )       | >2000 mg/kg           | >22 mg/L ( Rat ) 1 h<br>>5.5 mg/L ( Rat ) 4 h |
| Sodium carbonate         | 2800 mg/kg ( Rat )       | > 2000 mg/kg (rabbit) | 2.3 mg/l 2h (Rat)                             |
| Bismuth citrate          |                          |                       | LC50 > 5.07 mg/L ( Rat ) 4 h                  |

#### Chronic Toxicity

#### Carcinogenicity

There are no known carcinogenic chemicals in this product

#### Legend:

X - Listed 'I' - Not Listed PMN - Indicates a commenced PMN substance XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B))

#### Sensitization

No information available

#### Mutagenic Effects

No information available

#### Reproductive Effects

No information available

#### Developmental Effects

No information available

#### Target Organs

No information available.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component          | Freshwater Fish                                                                                            | Water Flea                                                                                  | Freshwater Algae | Microtox |
|--------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------|----------|
| Ferrous sulfate    | LC50: = 925 mg/L, 96h static (Poecilia reticulata)<br>LC50: = 0.56 mg/L, 96h semi-static (Cyprinus carpio) | EC50: = 152 mg/L, 48h (Daphnia magna)<br>EC50: 6.15 - 9.26 mg/L, 48h Static (Daphnia magna) |                  |          |
| Ammonium hydroxide | 0.53 mg/l LC50 96h<br>0.75 - 3.4 mg/l LC50                                                                 | EC50: 0.66 mg/L/48h                                                                         | -                | -        |

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|                  |                                                                                       |                                          |  |                      |
|------------------|---------------------------------------------------------------------------------------|------------------------------------------|--|----------------------|
|                  | 96h<br>8.2 mg/L LC50 96h                                                              |                                          |  |                      |
| Sodium sulfite   |                                                                                       |                                          |  | EC50 = 770 mg/L 17 h |
| Sodium carbonate | Lepomis macrochirus:<br>LC50: 300 mg/L/96h<br>Gambusia affinis: LC50:<br>740 mg/L/96h | EC50: = 265 mg/L, 48h<br>(Daphnia magna) |  | -                    |

## Persistence and degradability

### Degradation in sewage treatment plant

No information available

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

No information available

| Component          | log Pow | Bioconcentration factor (BCF) |
|--------------------|---------|-------------------------------|
| C.I. Basic green 1 | 0.765   | No data available             |
| Sodium sulfite     | -4      | No data available             |

## Mobility in soil

No information available.

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

#### Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed

| Component          | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Ferrous sulfate    | -         | X    | X   | X     | X    | X    | X     | X    | KE-21121 |
| Ammonium hydroxide | 215-647-6 | X    | X   | X     | X    | X    | X     | X    | KE-01688 |

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|                             |           |   |   |   |   |   |   |   |          |
|-----------------------------|-----------|---|---|---|---|---|---|---|----------|
| C.I. Basic green 1          | -         | X | X | X | X | X | X | X | KE-06975 |
| Peptones, connective tissue | 310-118-7 | - | - | - | - | - | - | - | KE-28132 |
| Glucose                     | -         | X | X | X | X | X | X | X | KE-17727 |
| Agar                        | -         | X | X | X | - | - | X | X | KE-00275 |
| Sodium phosphate dibasic    | -         | X | X | X | X | X | X | X | KE-12344 |
| Sodium sulfite              | -         | X | X | X | X | X | X | X | KE-31612 |
| Sodium carbonate            | -         | X | X | X | X | X | X | X | KE-31380 |
| Ferric ammonium citrate     | 214-686-6 | X | X | X | - | - | X | X | KE-01694 |
| Bismuth citrate             | -         | X | - | - | X | X | X | X | -        |

## National Regulations

### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

| Component       | Persistent Organic Pollutant | Ozone Depletion Potential | Pesticides Act 1974 |
|-----------------|------------------------------|---------------------------|---------------------|
| Ferrous sulfate |                              |                           | X                   |

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

INECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
PICCS - Philippines Inventory of Chemicals and Chemical Substances  
IECSC - Chinese Inventory of Existing Chemical Substances  
KECL - Korean Existing and Evaluated Chemical Substances

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**



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## Disclaimer

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**End of Safety Data Sheet**